Can Optimal Decision Trees Achieve Competitive Sentiment Classification While Providing Substantially Better Interpretability Than Transformer-Based Models?

An empirical comparison of optimal decision trees, logistic regression, and DistilBERT on SST-2

Arian Hosseini
University of Tehran

Abstract

We investigate whether optimal decision trees — trees fit by exact dynamic-programming search rather than greedy heuristics — can approach the accuracy of standard sentiment classifiers while remaining fully transparent. Using the Stanford Sentiment Treebank (SST-2), we train a Logistic Regression baseline, three variants of an optimal decision tree classifier (STreeD, via the PySTREED package) fit on hand-crafted linguistic features, reduced TF-IDF features, and a hybrid of the two, and a fine-tuned DistilBERT transformer. DistilBERT reaches 93.0% test accuracy, Logistic Regression reaches 81.1%, and the best optimal tree (hybrid features, depth 4) reaches 70.2% using only 16 human-readable rules. We show that the accuracy gap between trees and the transformer concentrates specifically in sentences with contrastive syntactic structure (e.g. "fun, but it grows tedious"), and that combining lexical and linguistic features gives optimal trees a small but consistent edge over either feature type alone. We conclude that optimal decision trees are not competitive with transformers on raw accuracy for this task, but the small, exact rule sets they produce make the accuracy–interpretability trade-off explicit and quantifiable in a way that post-hoc explanation methods for transformers cannot match.

1. Introduction

Sentiment classification is one of the most thoroughly benchmarked tasks in natural language processing, and modern transformer-based models routinely exceed 90% accuracy on standard datasets. That accuracy comes at a cost: the internal decision process of a fine-tuned transformer is not directly inspectable, and practitioners who need to justify individual predictions — in content moderation, market research, or any regulated setting — must rely on post-hoc explanation tools such as SHAP or LIME, which approximate a model's behavior locally rather than describing it exactly.

Decision trees offer the opposite trade-off: a shallow tree is exactly and globally interpretable, since every prediction can be traced through a short sequence of human-readable conditions. Classical decision tree induction (e.g. CART) builds trees greedily, one split at a time, which can produce trees that are locally sensible but globally suboptimal. Optimal decision tree methods instead search for the tree that provably maximizes accuracy for a given depth, using dynamic programming with bounds to make the search tractable at realistic dataset sizes (van der Linden et al., 2023). This raises a natural question, which is the central question of this paper:

Can optimal decision trees achieve competitive sentiment classification while providing substantially better interpretability than transformer-based models?

To answer it, we build three classifiers that sit at different points on the accuracy–interpretability spectrum — a bag-of-words Logistic Regression baseline, an optimal decision tree (STreeD) trained on three different feature representations, and a fine-tuned DistilBERT transformer — and evaluate all three on the same held-out SST-2 test set. Beyond raw accuracy, we compare the three models along the dimensions the plan for this project specifically called out: model complexity (tree depth, number of rules), computational efficiency, the form each model's explanations take, and the specific linguistic phenomena each model does and does not handle correctly. Our main empirical findings are:

- Optimal trees trade roughly 11–23 accuracy points for full interpretability, depending on the feature set.
- Aggregated linguistic features (e.g. a sentence-level sentiment score) are a far more efficient use of a shallow tree's limited depth budget than individual sparse TF-IDF terms.
- Combining linguistic and lexical features gives a small, consistent improvement over either alone, and — unlike the linguistic-only tree — continues to improve as depth increases.
- The transformer's advantage over both interpretable models is concentrated in sentences with contrastive conjunctions ("but", "though", "however"), which require resolving how two opposing sentiment cues combine — something no feature set used here represents directly.

2. Related Work

SST-2 is the binary-sentiment subset of the Stanford Sentiment Treebank (Socher et al., 2013) and is one of the nine tasks in the GLUE benchmark (Wang et al., 2018). It has been used to evaluate models ranging from simple bag-of-words classifiers to large pretrained transformers, which makes it a convenient common ground for comparing models of very different character.

Optimal decision tree induction has moved from a theoretical curiosity to a practical tool over the past decade, as dynamic-programming methods with tight bounds have made exact search feasible on datasets with tens of thousands of instances (van der Linden et al., 2023). We use STreeD specifically because, unlike many optimal-tree methods restricted to a single accuracy objective, it is designed to scale to the dataset sizes and feature counts typical of text classification while still returning a provably optimal tree for a given depth and binarization.

On the transformer side, we use DistilBERT (Sanh et al., 2019), a distilled version of BERT (Devlin et al., 2019) that retains roughly 97% of BERT's language understanding performance while being 40% smaller and 60% faster, making it a practical choice for a project with limited compute. For local explanations of DistilBERT's predictions, we use SHAP (Lundberg & Lee, 2017), which attributes each prediction to the individual input tokens using a game-theoretic notion of feature contribution.

3. Dataset

We use the official SST-2 train/validation/test split. Unlike the blind GLUE leaderboard test set, the split used here retains gold labels for all three partitions (a common and appropriate substitution for offline model comparison, since we do not submit to the GLUE leaderboard), giving 67,349 training sentences, 872 validation sentences, and 1,821 labeled test sentences. Class balance is close to even in the training set (55.8% positive) and almost exactly even in the test set (49.9% positive), so a majority-class baseline achieves only 50.1% test accuracy.

SST-2 sentences are short, single-clause snippets extracted from Rotten Tomatoes movie reviews and normalized (lowercased, tokenized) as part of the original Stanford Sentiment Treebank release. This normalization has a direct consequence for feature engineering: capitalization-based surface features (e.g. counting ALL-CAPS words) are uninformative on this corpus because no capitalized emphasis survives the normalization — we discovered this directly when an all-caps-word count feature came back constant at zero across all 67,349 training sentences, and dropped it from the feature set accordingly.

4. Methodology

4.1 Preprocessing

We used two preprocessing pipelines, matched to what each downstream model expects. For DistilBERT, sentences are passed through the model's own WordPiece tokenizer with minimal intervention — truncation and padding to a maximum length of 64 tokens, which comfortably covers SST-2's short sentences. For the classical pipeline (Logistic Regression and all PySTREED variants), sentences are lowercased and whitespace-normalized; punctuation is retained, since punctuation (particularly exclamation marks and commas) is used directly as a surface feature and as a signal within the TF-IDF vocabulary.

4.2 Feature Engineering

We compared three feature representations for the classical models, following the project's original design:

Feature Set 1 — TF-IDF. A 5,000-dimensional TF-IDF vector over unigrams and bigrams (minimum document frequency of 2), used as the sole representation for the Logistic Regression baseline.

Feature Set 2 — Linguistic. 14 hand-crafted features spanning five categories: lexical (AFINN-based positive/negative word counts and sentence-level sentiment score; Nielsen, 2011), syntax (adjective, adverb, verb, and noun counts via spaCy POS tagging), negation (count of not/never/no/n't/etc.), intensity (count of very/really/extremely/etc.), and surface form (sentence length, average word length, exclamation/question mark counts, comma count).

Feature Set 3 — Hybrid. The 14 linguistic features concatenated with the 200 TF-IDF terms most associated with the sentiment label by a chi-squared test, for 214 features total. Following standard practice for keeping optimal-tree search tractable, we deliberately did not feed the full 5,000-term TF-IDF vocabulary into PySTREED — doing so would make the resulting trees far less compact and readable, defeating the purpose of using a tree at all.

4.3 Models

Model 1 — Logistic Regression. Trained on the full 5,000-dimensional TF-IDF representation (scikit-learn, L2 regularization, C = 1.0). This serves as a strong, standard, but only moderately interpretable baseline — its decision is a weighted sum over 5,000 coefficients, which is inspectable in aggregate (e.g. top weighted terms) but not reducible to a short rule.

Model 2 — PySTREED (optimal decision tree). We trained STreeD classifiers at depths 1 through 4 (and additionally depth 5 for the linguistic feature set) on each of the three feature sets described above, for nine to ten total tree fits. STreeD searches for the tree that provably maximizes training accuracy for a given maximum depth using dynamic programming with bounds, rather than the greedy, locally-optimal splits used by CART-style trees.

Model 3 — DistilBERT. distilbert-base-uncased (66.96M parameters) fine-tuned for 3 epochs (batch size 32, learning rate 2e-5) directly on raw sentence text, with no hand-crafted features. This is the accuracy benchmark and the model with no directly readable global decision rule.

4.4 Evaluation Protocol

All models were trained on the same 67,349-sentence training split and evaluated on the same 872-sentence validation split and 1,821-sentence test split, so that accuracy, complexity, and error-pattern comparisons are made on identical data throughout. We report accuracy, precision, recall, and F1 on both validation and test sets, plus training time and per-instance inference time as efficiency measures. For PySTREED, we additionally report tree depth, number of leaves (equivalently, number of decision rules), and average rule length (the mean number of conditions from root to leaf) as complexity measures.

5. Results

5.1 Overall Performance

Table 1 reports validation and test performance for every model. DistilBERT is the clear accuracy leader at 93.0% test accuracy, roughly 12 points ahead of Logistic Regression and 23 points ahead of the best optimal tree. Among the three PySTREED variants, the hybrid feature set is consistently best, the linguistic-only feature set is a close second, and the reduced-TF-IDF feature set is a distant third — at its best, barely 5 points above the majority-class baseline.

Model	Features	Test Acc.	Test Prec.	Test Rec.	Test F1	Complexity
Majority baseline	—	50.1%	—	—	—	trivial
PySTREED	TF-IDF reduced (200)	55.3%	52.9%	94.1%	67.7%	15 leaves, depth 4
PySTREED	Linguistic (14)	68.0%	64.2%	81.1%	71.7%	8 leaves, depth 3
PySTREED	Hybrid (214)	70.2%	66.9%	79.9%	72.8%	16 leaves, depth 4
Logistic Regression	TF-IDF (5000)	81.1%	79.0%	84.5%	81.7%	5000 weights
DistilBERT	raw text	93.0%	91.3%	94.9%	93.1%	66.96M params

Table 1. Test-set performance and model complexity. PySTREED rows show each feature set's best-performing depth.

5.2 The Complexity–Accuracy Trade-off

Figure 1 tracks test accuracy against tree depth for all three PySTREED feature sets. Two patterns stand out. First, the linguistic-only tree plateaus almost immediately: depth 3 (68.0%) is essentially as good as depth 5 (67.5%), meaning the 14 hand-crafted features simply do not contain enough additional signal for extra depth to exploit. Second, the hybrid tree is the only variant that keeps improving through depth 4, suggesting the tree uses linguistic features for the coarse, high-level splits and individual TF-IDF terms to sharpen specific edge cases deeper in the tree — a pattern directly visible in the extracted rules (Section 6.2). The reduced-TF-IDF tree, by contrast, stays close to the majority baseline at every depth we tested: individual sparse word-presence features are simply too weak, on their own, to anchor the large, confident splits a shallow tree needs.

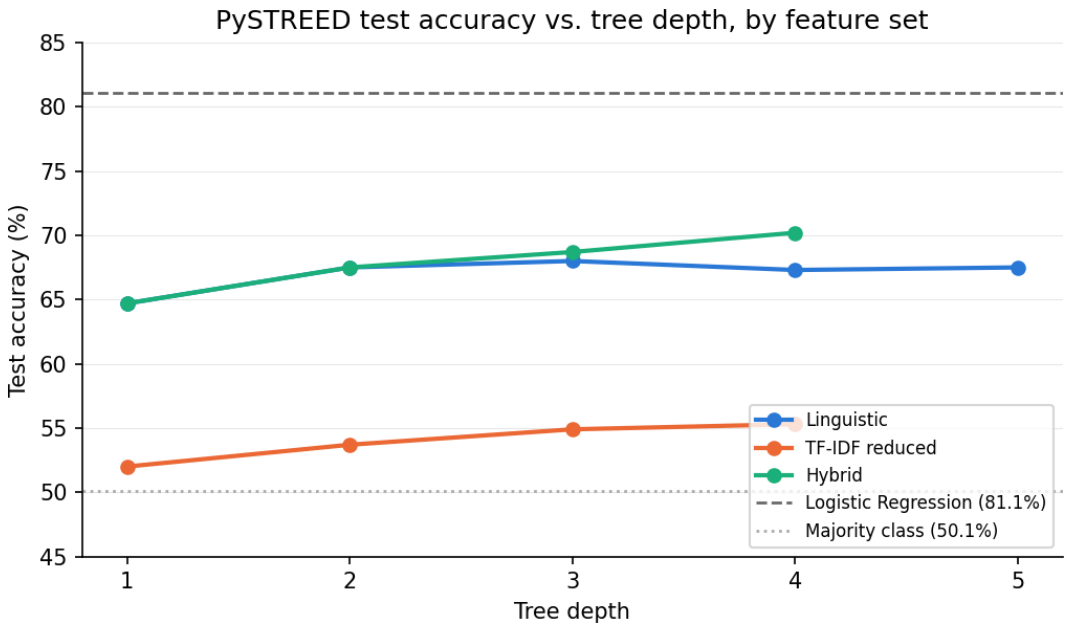


Figure 1. Test accuracy vs. tree depth, by feature set, with Logistic Regression and majority-class reference lines.

5.3 Efficiency

Table 2 reports training and inference time. All classical models were trained on a single CPU core; DistilBERT was fine-tuned on a T4 GPU, so training times are not directly comparable across hardware, but inference times are indicative of deployment cost on equivalent hardware. Logistic Regression and the linguistic-only tree are both essentially instantaneous at inference time; DistilBERT is roughly 300 times slower per instance than the best PySTREED tree, though still fast enough (1.7ms) for most practical applications.

Model	Train time	Inference (ms/instance)
Logistic Regression	0.42 s (CPU)	0.0002
PySTREED (linguistic, depth 3)	0.93 s (CPU)	0.0055
PySTREED (hybrid, depth 4)	120.8 s (CPU)	0.0396
DistilBERT	1143 s / 19.0 min (GPU)	1.71

Table 2. Training and inference cost. DistilBERT trained on GPU; all other models on CPU.

6. Interpretability Analysis

The three model families produce explanations of fundamentally different character: Logistic Regression yields global feature weights, PySTREED yields exact decision rules, and DistilBERT yields only local, post-hoc, per-instance explanations. This section examines each in turn.

6.1 Logistic Regression: Feature Weights

Logistic Regression's coefficients form a global, if unwieldy, explanation: every one of the 5,000 TF-IDF terms has a fixed weight, and the ten most positive and most negative weights (Table 3) are exactly the strongly polarized adjectives one would expect from a sentiment lexicon. This is reassuring — the model has learned recognizable sentiment vocabulary rather than a spurious correlation — but a 5,000-coefficient vector is not something a person can act on rule-by-rule the way they can with a tree; it is interpretable in aggregate, not per-decision.

Rank	Top positive terms (weight)	Top negative terms (weight)
1	powerful (+4.94)	bad (−5.56)
2	remarkable (+4.90)	lacks (−5.41)
3	hilarious (+4.77)	too (−5.36)
4	good (+4.75)	worst (−5.09)
5	fun (+4.74)	lacking (−4.99)
6	enjoyable (+4.64)	mess (−4.57)
7	beautiful (+4.43)	failure (−4.56)
8	entertaining (+4.34)	stupid (−4.55)
9	fascinating (+4.32)	not (−4.47)
10	solid (+4.20)	less (−4.46)

Table 3. Ten most positive and most negative Logistic Regression coefficients.

6.2 PySTREED: Exact Decision Rules

Unlike Logistic Regression, every PySTREED tree can be read as a small, complete, and exact set of IF/THEN rules — exact in the sense that the rules are not an approximation of the model, they are the model. We show two examples: the cleanest tree we trained (linguistic features, depth 3, 8 rules) and the best-performing tree overall (hybrid features, depth 4, 16 rules).

The depth-3 linguistic tree splits first on negation, then on a mix of sentence length, exclamation marks, and intensifier count, and finally on the AFINN sentiment score — a clean example of the tree combining a structural cue (negation) with a lexical polarity cue (AFINN score) exactly as a linguist would expect:

```
IF negation_count > 0.5 AND sentence_length > 18 AND afinn_score > 1.0  →  Positive
IF negation_count > 0.5 AND sentence_length > 18 AND afinn_score ≤ 1.0  →  Negative
IF negation_count > 0.5 AND sentence_length ≤ 18 AND no exclamation mark  →  Negative
IF negation_count > 0.5 AND sentence_length ≤ 18 AND has exclamation mark  →  Positive
IF negation_count ≤ 0.5 AND intensifier_count > 0.5 AND afinn_score > 0.0  →  Positive
IF negation_count ≤ 0.5 AND intensifier_count > 0.5 AND afinn_score ≤ 0.0  →  Negative
IF negation_count ≤ 0.5 AND intensifier_count ≤ 0.5 AND afinn_score > −1.0  →  Positive
IF negation_count ≤ 0.5 AND intensifier_count ≤ 0.5 AND afinn_score ≤ −1.0  →  Negative
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The depth-4 hybrid tree (16 rules; full listing in the accompanying supplementary file) follows the same top-level structure — negation, then AFINN score — but its deeper splits increasingly rely on individual TF-IDF terms (tfidf_film, tfidf_funny, tfidf_like, tfidf_compelling) rather than aggregate linguistic counts. This is the clearest evidence for why the hybrid tree keeps improving with depth where the linguistic-only tree does not: once the aggregate features have separated the "easy" cases, individual lexical items are what is left to resolve the remaining ambiguity, and only the hybrid feature set has them available.

6.3 DistilBERT: Local Explanations via SHAP

DistilBERT offers no equivalent global rule set; the closest available explanation is a per-instance SHAP attribution, which distributes a prediction across the input tokens. Table 4 shows SHAP attributions for four representative sentences. For straightforwardly worded sentences, SHAP recovers essentially the same sentiment vocabulary the Logistic Regression weights and the AFINN lexicon already surface ("loved", "wonderful", "stunning", "beautifully"). The more informative case is the sarcastic sentence “It's so bad that it's actually entertaining” — correctly classified as negative — where SHAP shows “bad” dominating the prediction (−0.641) despite “entertaining” pulling in the opposite direction (+0.279), which is a specific, checkable claim about how the model resolved the conflict. No tree or linear model in this study can produce that kind of per-instance, per-token account; it is also worth noting that a SHAP explanation describes one prediction at a time and does not, by itself, tell us what the model will do on the next sentence, unlike a PySTREED rule.

Sentence	Top contributing tokens (SHAP)
“This movie was absolutely wonderful and I loved every minute of it.”	loved (+0.22), wonderful (+0.17), absolutely (+0.11)
“It's so bad that it's actually entertaining.”	bad (−0.64), entertaining (+0.28), that (−0.06)
“This is not a good film, it was boring and dull.”	not (−0.31), dull (−0.12), boring (−0.08)
“A stunning, beautifully acted, deeply moving drama.”	deeply (+0.14), moving (+0.12), beautifully (+0.11)

Table 4. SHAP token attributions for four example sentences (positive value = pushes toward “positive”).

6.4 Summary: Three Kinds of Explanation

Model	Explanation type	Scope	Exact or approximate
Logistic Regression	Feature weights	Global	Exact
PySTREED	Decision rules	Global	Exact
DistilBERT	SHAP attribution	Local (per instance)	Approximate

Table 5. The three explanation types compared. Only PySTREED gives an explanation that is both global and exact.

7. Error Analysis

We compare error patterns on the shared 1,821-sentence test set, first between Logistic Regression and the best PySTREED tree (hybrid, depth 4), and then between the interpretable models and DistilBERT.

7.1 Logistic Regression vs. the Best Tree

Group	n	% of test set
Both correct	1118	61.4%
Only Logistic Regression correct	358	19.7%
Only tree correct	161	8.8%
Both wrong	184	10.1%

Table 6. Agreement between Logistic Regression and the best PySTREED tree (exhaustive partition of the test set).

Sentences only Logistic Regression gets right tend to use understated or backhanded negativity that does not trip any of the tree's explicit negation or AFINN thresholds — e.g. “a bravura exercise in emptiness” and “the angst-ridden, affluent slacker characters are more grating than engaging.” Neither phrase contains an AFINN-scored negative word or a negation token; Logistic Regression catches them because “emptiness” and “grating” happen to carry negative TF-IDF weight individually, a distinction the tree's 200-term reduced vocabulary does not always preserve. Sentences only the tree gets right typically involve negation interacting with an otherwise positive-sounding phrase, which the tree's explicit negation_count split is built to catch directly.

7.2 Interpretable Models vs. DistilBERT

Following the four-group scheme in the original project plan, we compare the best tree and Logistic Regression jointly against DistilBERT:

Group	n	% of test set	Description
Everyone correct	1085	59.6%	Tree, Logistic Regression, and DistilBERT all correct
Only DistilBERT correct	140	7.7%	Both classical models wrong
Tree correct, DistilBERT wrong	56	3.1%	Regardless of Logistic Regression
Everyone wrong	44	2.4%	All three models wrong

Table 7. Four-group error comparison (the remaining 27.2% of the test set falls into mixed patterns not covered by these four groups, e.g. only Logistic Regression correct).

The “only DistilBERT correct” group is where the transformer's advantage is concentrated, and it has a specific, checkable linguistic signature: sentences containing a contrastive conjunction (but, though, however, yet, despite) make up 23.6% of this group, compared to just 12.0% of the “everyone correct” group and 13.7% of the test set overall. Representative examples include “the fight scenes are fun, but it grows tedious” and “it delivers some chills and sustained unease, but flounders in its quest for deeper meaning” — both negative overall, despite containing an early positive-sounding clause. None of our linguistic features represent clause-level contrast directly; negation_count and afinn_score both operate at the sentence level and cannot capture that the second half of a sentence overrides the sentiment established by the first. This group's sentences are also modestly longer on average (20.9 words vs. 18.8

for the “everyone correct” group), consistent with the original plan's hypothesis that long-distance dependencies favor the transformer.

The “tree correct, DistilBERT wrong” group is smaller (3.1%) and less thematically consistent, but several examples share a pattern of heavily hedged or backhanded praise/criticism where the overall sentence-level polarity is arguably genuinely ambiguous even to a human reader — e.g. “one of those films that i wanted to like much more than i actually did” (negative) and “an impossible romance, but we root for the patronized iranian lad” (positive). We interpret this less as the tree being linguistically sophisticated and more as DistilBERT occasionally overweighting a strong local cue (e.g. “wanted to like”) that the simpler negation/AFINN-based tree does not key on as strongly.

The “everyone wrong” group (2.4%) is dominated by very short, terse fragments (“moot point.”) and sentences whose sentiment depends on real-world or franchise knowledge the models do not have access to, which we take as a plausible signal of residual label noise or genuine ambiguity in the original annotation, rather than a gap any of these three models could close with more data alone.

8. Linguistic Analysis of the Learned Rules

Reading the PySTREED trees as linguistic objects rather than purely as classifiers, three phenomena recur across nearly every tree we trained, at every depth:

Negation. negation_count is the root split (or immediately below it) in both the linguistic and hybrid trees at every depth we tested. This matches a well-established finding in sentiment analysis: negation is one of the single strongest and most reliable structural signals for sentiment, because it can invert the polarity an otherwise-positive lexical item would suggest.

Sentiment intensity / evaluation. Once negation has been resolved, every tree next splits on either afinn_score directly or on intensifier_count — i.e. having established whether a negation is present, the model asks how strong the (possibly negated) sentiment signal is. This two-stage structure (resolve polarity-flipping first, then magnitude) is a sensible and human-interpretable decomposition of the sentiment task, and it emerged from exact optimization rather than being designed in.

Lexical specificity at the leaves. In the hybrid tree specifically, the deepest splits are individual content words (funny, compelling, human, like) rather than aggregate counts — the tree uses broad linguistic categories to do the coarse sorting and falls back on specific vocabulary only once the aggregate signal is ambiguous. This mirrors, at a much smaller scale, how the Logistic Regression and DistilBERT SHAP explanations both ultimately bottom out in specific lexical items as well.

Two limitations of the rule sets are equally worth noting. First, not every split is semantically clean: several deeper leaves in the depth-4 trees combine conditions (e.g. comma_count and afinn_score together) in ways that are statistically justified by the training data but do not correspond to any obvious linguistic generalization — a reminder that depth-limited optimal accuracy and human-interpretable structure are related but not identical goals. Second, no feature set used here represents clause-level contrast (Section 7.2), which is precisely the phenomenon behind the transformer's largest accuracy advantage; a natural extension would be a feature counting contrastive conjunctions and their position within the sentence.

9. Discussion

Returning to the central research question: optimal decision trees do not achieve competitive accuracy with a fine-tuned transformer on SST-2 — a 23-point gap at their best (hybrid, depth 4) is substantial by any standard. They do, however, close roughly a third of the way from a majority-class baseline to Logistic Regression using only 14–214 features and 8–16 rules, and unlike Logistic Regression's 5,000 coefficients or DistilBERT's 67 million parameters and post-hoc SHAP attributions, every one of those rules is both exact and small enough to read in full. Whether that trade-off is worth making depends entirely on the application: for a deployment where every prediction must be individually justifiable in plain language — content moderation appeals, automated grading feedback, clinical or financial decision support — a 68–70% accurate model whose entire logic fits on one page may be preferable to a 93% accurate model whose logic cannot be fully stated at all, particularly if the interpretable model can be layered with a fallback for low-confidence cases.

The more actionable finding, though, is that feature choice determines how much of the accuracy gap is closable within the interpretable-model family at all. TF-IDF alone is a poor fit for shallow optimal trees, not because the information is absent but because sparse single-term features cannot anchor confident, generalizable splits the way aggregate linguistic scores can. Combining a small number of linguistic aggregates with a heavily reduced lexical vocabulary recovers most of the tree's achievable accuracy while keeping the rule set compact — a design choice that matters more, in our results, than swapping in a more powerful tree-learning algorithm would.

10. Limitations

- Single random seed and no cross-validation: reported numbers are point estimates from one train/test split each, without confidence intervals, so small accuracy differences (e.g. between hybrid depths 3 and 4) should be read cautiously.
- Restricted depth range (1–5): deeper optimal trees were not explored due to the super-linear growth in STreeD's search time with depth (hybrid depth 4 alone took over two minutes on a single CPU core); it remains possible that greater depth would narrow the accuracy gap further, at the cost of interpretability.
- Chi-squared feature selection for the reduced TF-IDF and hybrid sets is a simple filter method; a more sophisticated selection procedure might substantially change the reduced-TF-IDF tree's weak performance.
- SST-2 sentences are short, single-clause, pre-normalized movie-review snippets; findings (particularly the dead all-caps feature and the sentence-length effect) may not transfer to longer, less curated text such as full reviews or social media posts.
- DistilBERT was fine-tuned for a single run of 3 epochs with fixed hyperparameters; no hyperparameter search was performed, so its reported accuracy is a reasonable but not necessarily maximal estimate of what the model can achieve on this task.
- The linguistic feature set (14 features) is a design choice, not an exhaustive account of sentiment-relevant linguistic structure; the clause-level contrast phenomenon identified in Section 7.2 is a concrete, testable gap in it.

11. Conclusion

We compared Logistic Regression, optimal decision trees (PySTREED) on three feature representations, and DistilBERT on SST-2 sentiment classification, evaluating not just accuracy but rule complexity, efficiency, explanation type, and specific error patterns. Optimal trees are not accuracy-competitive with a fine-tuned transformer, but they offer something no amount of post-hoc explanation can: a small, exact, global rule set, and our results show that careful feature engineering — specifically, combining aggregate linguistic signals with a small set of individual lexical terms — narrows the achievable accuracy gap considerably more than depth alone does. The transformer's remaining advantage is not diffuse; it is concentrated in a specific, identifiable linguistic pattern (contrastive sentence structure) that none of the feature sets used here represent, which points to a concrete direction for future work on interpretable sentiment models.

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